

Element E: Application of STEM Principles and Practices

Principle 1: Torque Generation

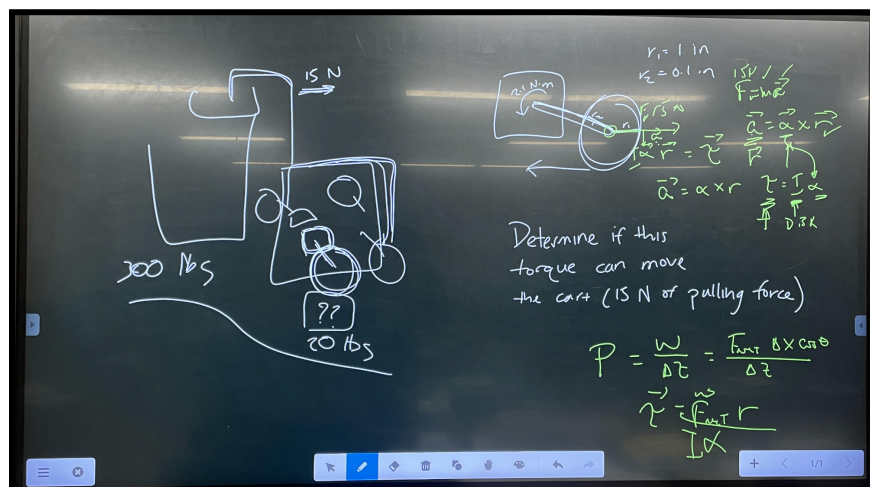
One of the biggest challenges of this design is to generate enough torque to move the cart and transfer it from the motors to the ground. We initially tried to calculate this torque with an equation resembling the following:

$$T_{required} = F_{required} * r_{wheel} * \mu\mu_s ???$$

where T is the torque rating of the motors, F is the force required to move the robot (determined experimentally) and r is the radius of the wheels, which is easy to change.

However, after consulting our physics teachers, Dr. Roberto Marrero and Mr. Mike Kiser, and a mechanical engineering student from Boston University, Mr. Jared Lawson, we determined that the actual mass of the chassis would play a role in this calculation.

Dr. Marrero was especially helpful here, as he helped walk us through the derivation seen below:



We approached the problem from a Newton's First Law perspective, where we knew the necessary force and the mass of the cart, meaning we only needed to solve for acceleration. Since linear acceleration is equal to angular acceleration times wheel radius, we then needed to solve for angular acceleration. Since torque (what we are solving for)

equals the moment of inertia of the wheel time angular acceleration, we can assemble the final equation:

$\bar{F} = m_T a$ $\bar{F} = m_T \alpha r$ $F = m_T \frac{\tau}{I} r$ $\tau = \frac{F I}{m r}$ $\therefore \tau = \frac{F m_w (a^2 + b^2)}{2 m_T r}$	<u>known eqs</u> $a = \alpha \cdot r$ $\tau = I \alpha$ $I_{\text{donut}} = \frac{1}{2} m_w (a^2 + b^2)$ a = hole radius b = wheel radius
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Where we solve for the torque required from the motor by the force required to move the cart (measurable), the mass of the wheels (can be designed for), the inner and outer radius of the wheels (can be designed for), and the total mass of the chassis (can be designed for).

Note that the original equation included the coefficient of static friction between the wheel and floor, while the new equation didn't. This is because we determined that the friction between the wheel and floor just had to exceed a certain threshold to allow the wheel to roll instead of slide.

Principle 2: Power Usage

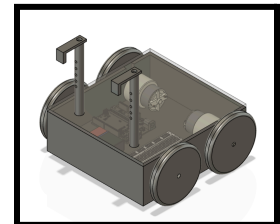
After finding how much torque our motors need to generate, we also have to calculate the size and performance of the battery required to power said motors. Once we identify motors with enough torque, we can calculate using:

$$mAh_{\text{required}} = 2 * I_{\text{motor}} * t_{\text{required}}$$

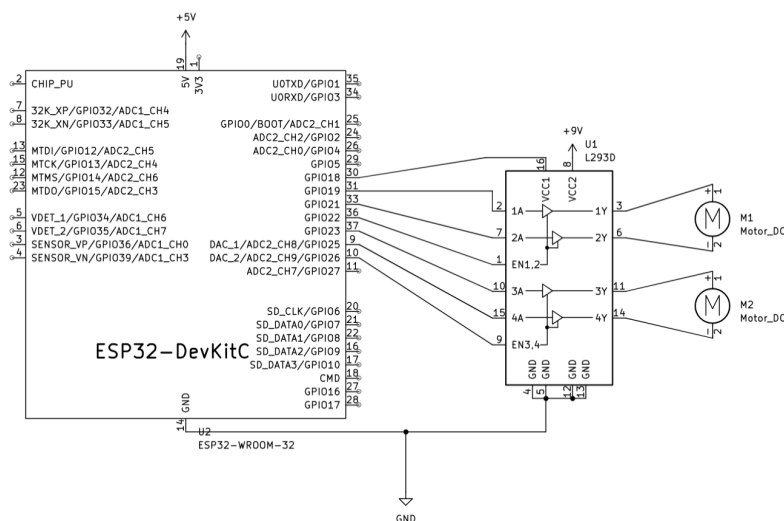
where mAh is the capacity of the battery, I is the peak current of a motor, and t is the time that the robot will be able to run for between each charge. The microcontroller and possibly speed controllers will also require some of this battery power but at a much lesser level than the motors.

Principle 3: Computer-Aided Design

As seen above, we used Fusion 360 to create a 3D design of our solution. This allowed us to present our ideas in a theoretical manner and get feedback from outside sources without having to spend time and money on a physical prototype. Due to the nature of the project, it's entirely possible that we could end up changing our chassis design, motors, batteries, and/or microcontroller.



Principle 4: Circuit Design



To understand how the electronic component of the solution would work, we designed and tested a basic electrical schematic in KiCad's schematic editor. The design integrates a WiFi-compatible microcontroller and an H-bridge integrated circuit to permit remote, precise control of the motors. Beyond providing a wiring plan for a physical prototype, this approach allowed us to use KiCad's native Electrical Rules Checker (ERC) to verify the continuity and functionality of our circuit.

KiCad's workflow between the schematic editor and PCB editor streamlines the design of printed circuit boards that integrate a microcontroller, motor controller, and motor outputs. This allows for product scalability and price reduction, removing the need to

handwire each component individually by opening the door to mass production of cheap PCBs.

Expert Review

As mentioned above, we worked with our physics teachers, Dr. Marrero and Mr. Kiser to solve some of the more complex physics problems that our design presents, as well as received input from Mr. Lawson, who recently completed his undergraduate degree in mechanical engineering.

Expert Signatures:

The image shows three handwritten signatures in black ink. The first signature on the left is 'Dr. Marrero', the middle one is 'Mr. Kiser', and the one on the right is 'Mr. Lawson'. They are all written in a cursive, flowing style.